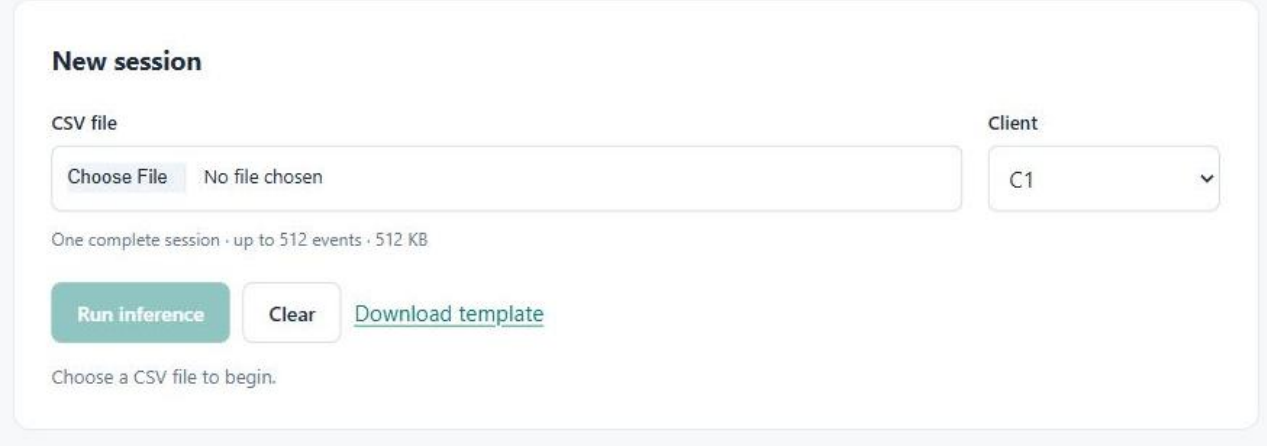
Session Check User Guide

Upload a session Run inference Review the result

Use this guide to test one complete event session, understand the score, record your review and download a result. The public website works independently of the research computer. You need a browser, Internet access and a CSV file; no installation or login is required.

Open Session Check

1 Start a session



1. Open the website. If the free host is waking up, allow about a minute. If Reconnect appears, select it.

2. For a first test, unzip Session_Check_Test_Samples.zip. Use the four CSV examples supplied with this guide.

3. Select Choose File under CSV file and choose 01_below_threshold.csv.

4. Select Client C2 to reproduce the below-threshold screenshot. For your own data, select the correct original client; the selector does not calibrate a new institution.

5. Select Run inference once. Wait for Inference complete. Controls are temporarily locked while processing.

6. Read the result, inspect Event contributions if needed, and save a review. Select Download result to keep a JSON record.

7. Select Clear to start again. It resets the file, result and review and selects C1. Changing the file or client also hides the old result; run inference again.

Use synthetic or appropriately de-identified data. The public website sends the CSV over HTTPS to the hosted server for processing.

Hosting note: the free service may sleep after inactivity and take about one minute to resume. Render documentation: https://render.com/docs/free

2 Understand the result

The anomaly score is the session's reconstruction error. Higher means poorer reconstruction under this detector. It is not a threat probability, is not a percentage and can exceed 1. The client threshold is the frozen 99th percentile of that client's routine-validation scores.

Score compared with threshold	Screen result	What to do
Strictly greater	Review recommended	Review the session and its context.
Less than or exactly equal	Below threshold	No alert under this rule. Investigate if other evidence warrants it.
Input or service error	No new result	Correct the issue and rerun; an error is not an alert.

Displayed numbers are rounded. The decision uses full precision. The 99th-percentile rule does not guarantee a 1% false-positive rate on new data.

Verified results for the two valid samples

Sample	Client	Anomaly score	Threshold	Result
01 below | 12 events	C1	0.1982352	0.2165964	Below threshold
01 below | 12 events	C2	0.1982352	0.2225361	Below threshold
01 below | 12 events	C3	0.1982352	0.2050999	Below threshold
02 above | 1 event	C1	1.707739	0.2165964	Review recommended
02 above | 1 event	C2	1.707739	0.2225361	Review recommended
02 above | 1 event	C3	1.707739	0.2050999	Review recommended

Sample 01 is below all three thresholds. For C2 it spans 01:00:00 to 01:08:15 UTC on 26 October 2026. Sample 02 exceeds all three thresholds. Its start and end are both 01:00:00 UTC because it contains one event.

The same trained model scores all clients, so the same input gives the same raw score; the threshold changes. These synthetic examples exercise the interface and have no verified threat labels. Their different lengths and scores do not establish a causal effect of session length.

Four possibilities when the real label is known

Model decision	Verified real activity	Interpretation
Review recommended	Threat	True positive
Review recommended	Legitimate	False positive
Below threshold	Threat	False negative
Below threshold	Legitimate	True negative

The score alone cannot identify which of these four cases applies. Neither a high score confirms a threat nor a low score proves safety.

3 Inspect and record your review

Read the event contributions

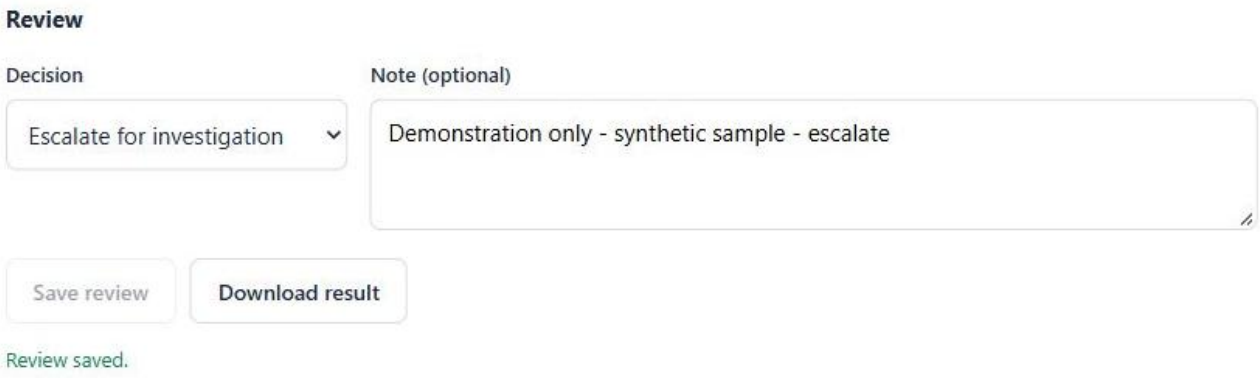
Open Event contributions. Highest contribution lists the largest score shares first; Sequence order restores positions 0, 1, 2 and so on. Each row shows the event ID, UTC time, department and resource.

Score share is the event's share of the reconstruction score, not its probability of being harmful. In sample 01, position 0 contributes 62.26%. In the one-event sample 02, that event contributes 100% by definition. Shares may differ slightly from a total of 100% because of rounding.

Choose a human review decision

Decision	Use when
Needs more context	You need more information before deciding.
Legitimate activity	Your review indicates an authorized explanation.
Escalate for investigation	You intend to refer the session for investigation.

Select a Decision, add a Note if useful, then select Save review. A blank note is allowed; the limit is 4,000 characters. Wait for Review saved. The Save review button then becomes disabled until you edit the decision or note.



Saving records your entry in the temporary browser session. It does not change the score, threshold or model. Escalate for investigation records a selection; it does not send a message, open a ticket, block access or perform an investigation.

Download and finish

Select Download result to save a JSON file. It contains the session, client, time window, full-precision scores, alert decision, event contributions, coverage and model details, plus the last saved review. Before review, human_review.status is pending for an alert and not_triggered below threshold.

If Unsaved changes appears, select Save review before downloading to include the new text. Otherwise the export contains the earlier saved review. Keep the downloaded file as your record; this page has no result history or JSON import.

4 Prepare your own CSV

Select Download template to obtain the exact header. It is a blank header template; add your own event rows. In Excel, export as CSV UTF-8, not an Excel workbook. Preserve the column names below.

Fields	Required columns
Session and event order	session_id, event_id, sequence_position, timestamp_utc
Categorical event details	action, event_category, department, role, resource_type, source_application, device_type, network_zone, location_type, purpose_of_use, relationship_type, outcome, ward_or_location
Boolean event details	assigned_patient, break_glass, on_shift

One file must contain one complete session with 1 to 512 events and at most 512 KiB (524,288 bytes; shown as 512 KB). Do not truncate or split a session merely to bypass the limit.

Use one nonempty session_id in every row. Give every event a nonempty, unique event_id.

Set sequence_position to 0, 1, 2, ... with no gaps or duplicates. Use plain integers, not 01 or 1.0. The engine sorts by this field, so shuffled file rows are accepted when positions remain correct.

Use a valid timestamp with an explicit UTC offset, for example 2026-10-26T01:00:00Z or 2026-10-26T01:00:00+00:00. Allow at most six fractional digits. Times cannot move backwards in sequence order; equal times are allowed.

Fill all three Boolean columns with true/false or 1/0. True/False and TRUE/FALSE are also accepted. Do not leave these cells blank.

Keep categorical spelling and case consistent with the source data. Leading or trailing spaces change the value. Do not enter the reserved tokens __UNK__ or __MISSING__.

Use only the template columns for the simplest workflow. If you add client or hospital_id, it must agree with the selected client. Duplicate column names and malformed CSV rows are rejected.

If some categories are unknown or blank

Ordinary unseen or blank categorical values can be accepted. The result then shows a coverage note because those cells are excluded as reconstruction targets. This changes the score calculation; inspect the source values before interpreting the result. Missing Boolean values and invalid timestamps are still rejected.

A controlled test with one unseen action and one blank department used 154 of 156 categorical targets and produced 0.1847772 for C1, below threshold. This lower score does not establish improved safety. Correct accidental missing values and rerun the complete session.

5 Test the examples and fix input errors

Run each file separately. Use C1 initially, then choose C2 or C3 and run again to compare thresholds. The two valid samples keep their raw score across clients, as shown on page 2.

Test file	Expected outcome	Explanation or correction
01_below_threshold.csv	Below threshold	12-event session; score 0.1982352.
02_above_threshold.csv	Review recommended	1-event session; score 1.707739.
03_invalid_sequence.csv	Rejected without a score	Positions begin at 1. Reindex 0 to N minus 1.
04_missing_columns.csv	Rejected without a score	Required columns are absent. Use the template header.

Input problems and solutions

Message or problem	Action
Run inference is disabled	Choose a nonempty CSV; wait for connection or processing to finish.
Choose a .csv file / UTF-8 encoding error	Export as CSV UTF-8. Renaming an .xlsx file is insufficient.
File is empty / session must contain events	Add the header and at least one complete event row.
File too large / more than 512 events	Use a supported complete session within both limits.
Missing or duplicate CSV column names	Restore the exact template header and remove duplicates.
Required cells missing / extra cells / field too long	Check commas and quoted text. Remove accidentally pasted long content.
Submit exactly one nonempty session_id	Keep one session per file and fill its session ID in every row.
event_id values must be nonempty and unique	Fill missing IDs and remove or correct duplicate IDs.
sequence_position must be canonical, contiguous and start at zero	Use exactly 0, 1, ... N minus 1, without gaps, spaces or duplicates.
Timestamp error / negative elapsed time	Correct the date, UTC offset and event order.
Missing or invalid Boolean	Fill assigned_patient, break_glass and on_shift with accepted values.
Raw literal conflicts with reserved token	Use the original category value or a genuine blank categorical cell.
CSV client or hospital_id disagrees	Select the correct client or correct the optional field.
Coverage note but inference completes	Check unknown or blank categories. The note is not a threat verdict.

After correcting a CSV, select the corrected file again and select Run inference. A rejected request has no valid new score and should not be interpreted as either model decision.

6 Recover and keep your results

Status or problem	What to do
Running or Processing session	Wait. Avoid repeated clicks; controls unlock after completion.
Server unavailable or cannot connect	Check Internet access, allow the host to wake, then select Reconnect.
Your session expired	Select Reconnect. The loaded CSV remains selected; run inference again.
Result not found in your session	Rerun inference and save the review again.
The model or service is busy	Wait briefly, then retry. Reconnect if offered.
Too many inference requests	Wait one minute before trying again.
Upload timed out or unreadable server response	Check the connection and retry. Refresh if the problem persists.
Request could not be processed	Do not use it as a result. Retry once; contact the maintainer if repeated.
Review was not saved	Copy your note elsewhere, resolve the displayed error and save again.
Save review is disabled	The review is already saved, another operation is running, or Reconnect is required.
Note is too long	Shorten it to 4,000 characters or fewer, then save.
Download does not contain your latest edit	Save review, wait for Review saved, then download again.

Keep a record before leaving

The public app keeps results and reviews temporarily in server memory, with at most three retained results per visitor. Browser sessions expire after about 20 minutes without API activity; capacity limits, cookie expiry or server restarts may require an earlier rerun. Download promptly. Refreshing or closing the page clears the displayed workflow, and earlier results cannot be reopened through this interface.

Clear resets the browser workflow; it does not request immediate server deletion. An already downloaded JSON remains on your computer. The separate local and portable versions save local files; this guide describes the public website.

Model and verification

This website uses the original trained FedAvg with SecAgg+ model C, Reference condition, seed 11, selected round 30. It performs inference only. Uploads and review decisions do not retrain or recalibrate it; SecAgg+ does not run during inference.

Verified on 26 October 2026: 19 browser workflow checks and 39 backend, boundary and contract checks passed. Browser checks covered upload, both results, client changes, both event orders, all review choices, saved and pending exports, unsaved edits, Clear, input errors and controlled expiry with Reconnect. All six valid sample and client combinations were scored in an isolated copy of the public backend.

Exact threshold equality was tested through the decision function. Busy handling and some exceptional service failures were checked against the implementation rather than forced on the public service. The examples demonstrate supported behavior; they do not establish detection accuracy on new data.